

Experiimnet:16

Develop a C program for implementing random access file for processing the employee details

Code:-

```
#include <stdio.h>
#include <stdlib.h>
struct Employee
{
    int empId;
    char empName[50];
    float empSalary;
};
int main()
{
    FILE *filePtr;
    struct Employee emp;
    int choice;
    filePtr = fopen("employee.dat", "rb+");
    if (filePtr == NULL)
    {
        filePtr = fopen("employee.dat", "wb+");

        if (filePtr == NULL)
        {
            printf("Error creating the file.\n");
            return 1;
        }
    }
}
```

```
do
{
    printf("\nEmployee Database Menu\n");
    printf("1. Add Employee\n");
    printf("2. Display Employee Details\n");
    printf("3. Update Employee Details\n");
    printf("4. Exit\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
    case 1:
        printf("Enter Employee ID: ");
        scanf("%d", &emp.empId);

        printf("Enter Employee Name: ");
        scanf("%s", emp.empName);
        printf("Enter Employee Salary: ");
        scanf("%f", &emp.empSalary);

        fseek(filePtr,
              (emp.empId - 1) * sizeof(struct Employee),
              SEEK_SET);
```

```
fwrite(&emp, sizeof(struct Employee), 1, filePtr);
```

```
printf("Employee details added successfully.\n");
```

```
break;
```

case 2:

```
printf("Enter Employee ID to display: ");
```

```
scanf("%d", &emp.empId);
```

```
fseek(filePtr,
```

```
    (emp.empId - 1) * sizeof(struct Employee),
```

```
    SEEK_SET);
```

```
fread(&emp, sizeof(struct Employee), 1, filePtr);
```

```
printf("\nEmployee ID    : %d\n", emp.empId);
```

```
printf("Employee Name   : %s\n", emp.empName);
```

```
printf("Employee Salary : %.2f\n", emp.empSalary);
```

```
break;
```

case 3:

```
printf("Enter Employee ID to update: ");
```

```
scanf("%d", &emp.empId);
```

```
fseek(filePtr,
```

```
    (emp.empId - 1) * sizeof(struct Employee),
```

```
    SEEK_SET);
```

```
fread(&emp, sizeof(struct Employee), 1, filePtr);

printf("Enter New Employee Name: ");
scanf("%s", emp.empName);

printf("Enter New Employee Salary: ");
scanf("%f", &emp.empSalary);

fseek(filePtr,
      (emp.empId - 1) * sizeof(struct Employee),
      SEEK_SET);

fwrite(&emp, sizeof(struct Employee), 1, filePtr);

printf("Employee details updated successfully.\n");
break;

case 4:
    printf("Exiting...\n");
    break;

default:
    printf("Invalid choice. Please try again.\n");
}

} while (choice != 4);
```

```
fclose(filePtr);
```

```
return 0;
```

```
}
```

Output:

```
Enter your choice: 5
Invalid choice. Please try again.

Employee Database Menu
1. Add Employee
2. Display Employee Details
3. Update Employee Details
4. Exit
Enter your choice: 6
Invalid choice. Please try again.

Employee Database Menu
1. Add Employee
2. Display Employee Details
3. Update Employee Details
4. Exit
Enter your choice: 6
Invalid choice. Please try again.

Employee Database Menu
1. Add Employee
2. Display Employee Details
3. Update Employee Details
4. Exit
Enter your choice: 4
Exiting...

...Program finished with exit code 0
Press ENTER to exit console.
```